

VAISHALI SONKER

AI Intern | Machine Learning | Data Analytics
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Professional Summary

Data Science undergraduate with hands-on experience in end-to-end machine learning pipelines, predictive modeling, and data analytics. Proficient in Python, Scikit-learn, XGBoost, SQL, and data visualization tools. Delivered ML models with 10–15% accuracy improvements and interactive dashboards for real-world business forecasting. Seeking an AI/ML Intern or Data Scientist role to drive data-driven product decisions.

Technical Skills

Languages: Python, SQL, JavaScript, NumPy, Pandas

ML / AI: Scikit-learn, XGBoost, SARIMA, Regression, Classification, RAG (Retrieval Augmented Generation), Semantic Search

Deep Learning: CNNs, RNNs, LSTM (Foundational), Transformers, Sentence Embeddings

Computer Vision: OpenCV, Image Classification, Defect Detection

Data Engineering: ETL Pipelines, Feature Engineering, EDA, Data Cleaning, Preprocessing

Databases: MySQL, PostgreSQL, MongoDB, FAISS (Vector Database)

Tools & Platforms: FastAPI, Jupyter Notebook, Streamlit, Git, Power BI, Excel (Advanced)

Experience

Machine Learning Intern | CNE Associates

Dec 2025 – Jan 2026

- Developed and evaluated classification and regression ML models using Scikit-learn, improving model accuracy by 10–15% through iterative feature engineering and hyperparameter tuning.
- Built end-to-end ML pipelines covering data ingestion, preprocessing, model training, and evaluation using RMSE and F1-score; reduced data inconsistencies by ~20% across 3+ datasets.
- Delivered reproducible model experiments with versioning notes and basic deployment workflows, shortening iteration cycles for the team.

Key Projects

Chat with PDFs — RAG-Based Document Q&A System | Python, FastAPI, FAISS, SentenceTransformers, Gemini API, PyMuPDF

- Built an end-to-end Retrieval Augmented Generation (RAG) pipeline from scratch — PDF ingestion, semantic chunking, embedding generation, FAISS vector search, and LLM answer generation — without high-level abstractions like LangChain.
- Reduced LLM token consumption by ~98% vs. naive full-document approach by retrieving only the top-5 semantically relevant chunks per query using all-MiniLM-L6-v2 (384-dim embeddings) and FAISS IndexFlatL2 similarity search.
- Implemented real-time streaming responses via FastAPI `StreamingResponse` and the browser Fetch API, replicating a ChatGPT-like UX with token-by-token output.
- Engineered prompts with context injection and hallucination-reduction constraints; persisted FAISS index to disk to maintain vector store state across server restarts.

Sales Forecasting Dashboard (Time Series + ML) | GitHub | Python, SARIMA, XGBoost, Streamlit, Pandas

- Reduced forecast RMSE by 18% by replacing SARIMA with XGBoost on 4+ years of time-series data after comparative evaluation using MAE and RMSE metrics.
- Engineered lag features, rolling averages, and date-based encodings to capture seasonality, improving prediction accuracy by 15% over the naive baseline.
- Deployed interactive Streamlit dashboard with 30–60 day business forecasts supporting inventory planning and demand estimation decisions.

Sales KPI Dashboard (Power BI) | GitHub | Power BI, Excel, Power Query

- Built Power BI dashboard tracking revenue, profit, and regional KPIs; cut manual reporting effort by ~40% using dynamic filters and drill-down visualizations.
- Cleaned and unified data from 5+ regional sources using Power Query, ensuring cross-regional consistency for accurate business analysis.

Education

Bachelor of Technology (B.Tech) — Computer Science Engineering | CPI: 7.19/10

Expected: 2027

3rd Year Undergraduate